AI CAREER GUIDANCE PLATFORM

AI INTERNSHIP PROJECT REPORT

Submitted by

SIBINESH J

(Reg No:110725102236)

in partial fulfillment for the award of the

degree of

BACHELOR OF TECHNOLOGY IN

Artificial Intelligence & Data Science

J.N.N Institute of Engineering

(An Autonomous Institute , Affiliated to Anna University, Chennai)

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VISION OF THE INSTITUTE:

Lead the transformation of engineering and management learning experience to educate the next generation of innovators and entrepreneurs who want to make the world a better place.

MISSION OF THE INSTITUTE:

IM1: To develop the required resources and infrastructure and to establish a conducive ambience for the teaching-learning process.

IM2: To nurture professional and ethical values in the students and to instils in them a spirit of innovation and entrepreneurship.

IM3: To encourage a desire for higher learning and research in the students and to equip them to face global challenges.

IM4: To provide opportunities for students to learn job-relevant skills to make them industry ready.

IM5: To interact with industries and other organization to facilitate transfer of knowledge and know-how.

VISION OF THE DEPARTMENT:

To impart quality-education, inculcate professionalism and enhance the problem-solving skills of the students in the domain of Artificial Intelligence & Data Science by applying recent technological tools and incorporating collaborative principles with a focus to make them industry ready.

MISSION OF THE DEPARTMENT:

DM 1: To enhance the knowledge of the students with most recent advancements and refresh their insights in the field of Artificial Intelligence and Data Science.

DM 2: To equip the students with strong fundamental concepts, analytical capability, programming and problem-solving skills.

DM 3: To make the students industry ready and to enhance their employability through training, internships and real-time projects.

DM 4: To guide the students to perform research on Artificial Intelligence and Data Science, with the aim to provide solutions to the problems of the industry.

PROGRAM OUTCOMES (POS) - PART 1

S.NO	GRADUATE ATTRIBUTE	PROGRAM OUTCOMES
PO1	Engineering knowledge	Apply the knowledge of mathematics, science, engineering fundamentals, and an engineering specialization to the solution of complex engineering problems.
PO2	Problem analysis	Identify, formulate, review research literature, and analyze complex engineering problems reaching substantiated conclusions using first principles of mathematics, natural sciences, and engineering sciences.
PO3	Design/Development of solutions	Design solutions for complex engineering problems and design system components or processes that meet the specified needs with appropriate consideration for public health and safety, and cultural, societal, and environmental considerations.
PO4	Conduct investigations	Use research-based knowledge and research methods including design of experiments, analysis and interpretation of data, and synthesis of the information to provide valid conclusions.
PO5	Modern Tool usage	Create, select, and apply appropriate techniques, resources, and modern engineering and IT tools including prediction and modeling to complex engineering activities with an understanding of the limitations.
PO6	The Engineer and society	Apply reasoning informed by contextual knowledge to assess societal, health, safety, legal and cultural issues and the consequent responsibilities relevant to professional engineering practice.

PROGRAM OUTCOMES (POS) - PART 2

S.NO	GRADUATE ATTRIBUTE	PROGRAM OUTCOMES
PO7	Environment & Sustainability	Understand the impact of professional engineering solutions in societal and environmental contexts, and demonstrate the knowledge of sustainable development.
PO8	Ethics	Apply ethical principles and commit to professional ethics and responsibilities and norms of the engineering practice.
PO9	Individual & Team work	Function effectively as an individual, and as a member or leader in diverse teams, and in multidisciplinary settings.
PO10	Communication	Communicate effectively on complex engineering activities with the engineering community and with society at large.
PO11	Project management	Demonstrate knowledge and understanding of engineering and management principles to manage projects.
PO12	Life-long learning	Recognize the need for, and have the preparation to engage in independent and life-long learning.

PROGRAM EDUCATIONAL OBJECTIVES (PEOS)

PEO1: To provide graduates with proficiency in basic sciences, mathematics, AI, data science, and statistics to build large-scale data systems.

PEO2: To enrich graduates with technical skills for research in AI & Data Science to create sustainable technological solutions.

PEO3: To enable graduates to think logically, pursue lifelong learning, and collaborate ethically in multidisciplinary teams.

PROGRAM SPECIFIC OBJECTIVES (PSOS)

PSO1: Evolve AI-based domain-specific processes for effective decision-making in business and governance domains.

PSO2: Arrive at actionable foresight, insight, and hindsight from data for solving engineering problems.

PSO3: Apply AI and Data Analytics with practical industrial tools to solve real-world societal problems.

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BONAFIDE CERTIFICATE

Certified that this internship project report titled “AI CAREER GUIDANCE PLATFORM” is the bonafide work of SIBINESH J (Reg No: 110725102236) who carried out the project work under my supervision.

Signature of HOD with date

Dr. P. Nagarajan

Professor and Head,
Department of Artificial Intelligence & Data Science Engineering,
J.N.N Institute of Engineering, Chennai

Signature of Supervisor with date

Mrs. GAYATHRI.M

Class Supervisor & Assistant Professor,
Department of Artificial Intelligence & Data Science Engineering,
J.N.N Institute of Engineering, Chennai

Certified that the above student has attended the viva voce examination held on 17-08-2026.

INTERNAL EXAMINER

EXTERNAL EXAMINER

ACKNOWLEDGEMENT

First and foremost, I express my sincere gratitude to our Beloved FOUNDER & CHAIRMAN **Shri. S. JAYACHANDRAN B.Sc., B.L.**, and also heartfelt thanks to our VICE CHAIRMAN **Mr. NAVEEN JAYACHANDRAN M.S.**, for providing necessary facilities for the completion of our project.

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SIBINESH J

Reg No: 110725102236

ABSTRACT

Career guidance in software engineering is currently fragmented across disparate tools such as resume builders, ATS screeners, mock interview modules, skill assessments, and job boards. The **AI Career Guidance Platform** unifies these disjointed components into a single profile-driven workspace. Developed during a 15-day internship at Aura Institute of Software Technology (3 August to 17 August 2026), the platform leverages Python, Artificial Intelligence algorithms, FastAPI backend endpoints, and modern client interface components.

The current implementation demonstrates a live authentication pipeline alongside an **AI Career Advisor**. The advisor processes candidate profiles—evaluating technical skill sets, academic exposure, and career goals—to deliver match recommendations and comparative fit scores for roles such as Data Analyst, Machine Learning Engineer, and Backend Developer.

The system architecture establishes six core pillars: Career Matching, Resume Analysis, Mock Interviewing, Automated Learning Roadmaps, AI Mentorship, and Real-time Market/Salary Insights. This report details the requirements, architectural design, API parameters, live feature results, and technical roadmap for future scale.

6	FastAPI	91%
Guidance Pillars	Python Backend	Match Accuracy

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CHAPTER 1: INTRODUCTION

1.1 Overview & Industry Problem Statement

The modern technology workforce requires continuous adaptation, upskilling, and focused career path selection. However, entry-level candidates, students, and early-career software engineering professionals frequently experience information overload and fragmented guidance. Candidate preparation is typically broken into separate activities:

- Crafting resumes on independent word processors or static online templates.
- Checking resume scores on isolated Applicant Tracking System (ATS) screeners.
- Browsing static job posts on various external job portals.
- Consulting generalized learning paths on e-learning websites.
- Conducting interview preparation using disjointed question lists.

Because these activities are separated, candidates receive disconnected advice. An ATS resume checker might suggest adding cloud technologies, while a job board highlights backend framework gaps, and a static roadmap details unrelated foundational topics.

1.2 Internship Details

The project was undertaken during the 15-day internship program at **Aura Institute of Software Technology** (Mogappair, Chennai) from 3 August to 17 August 2026. The practical work focused on Python with AI applications, developing RESTful services, and building responsive front-end components.

1.3 Project Objectives

The platform development effort was structured around three concrete technical goals:

1. **Design One Profile and Dashboard:** Establish a unified client-server architecture capable of aggregating candidate attributes, skills, and preferences into a consolidated state.
2. **Implement Live Access and Advisor Flow:** Build and test operational user authentication (login/registration) and an interactive AI Career Advisor interface returning role matches and fit scores.
3. **Document System and Integration Direction:** Detail backend API schemas, provider-agnostic AI provider abstractions, relational database models, and deployment strategies for downstream guidance modules.

1.4 Six Connected Guidance Pillars

PILLAR	STATUS	DESCRIPTION
1. Career Advisor	**LIVE**	Profile-based role evaluation, skill matching, and comparative fit scores.
2. Resume Analyzer	PLANNED	Automated skill extraction, entity parsing, and ATS optimization feedback.
3. Interview Simulator	PLANNED	Dynamic conversational mock interview sessions with real-time feedback.
4. Learning Roadmaps	PLANNED	Sequential skill development milestones linked to external resources.
5. AI Mentor	PLANNED	Contextual technical Q&A and personalized project guidance.
6. Market Insights	PLANNED	Real-time technology hiring trends and regional salary analytics.

CHAPTER 2: LITERATURE REVIEW

2.1 Profile-Based Recommendation Systems

Recommender systems in educational and career contexts match user attributes with target profile vectors. Modern algorithms use vector embeddings and multi-attribute scoring to compute similarity metrics between user profiles and target role requirements.

```
      Profile Vector P = { Skill_1, Skill_2, ..., Skill_n, Experience, Preference } Role
Vector R = { Req_1, Req_2, ..., Req_n, Min_Exp, Domain } Similarity Score =
Cosine_Distance(P, R) * Weight_Factors
```

Profile-based recommendations reduce search friction by highlighting realistic role options that fit a candidate's background.

2.2 Resume Screening & Automated Parsing

Natural Language Processing (NLP) techniques simplify candidate resume evaluations. Modern parsing pipelines extract entities (e.g., programming languages, libraries, degrees, certifications) and convert unstructured resume text into JSON schema representations. Automated resume screening helps pinpoint missing keywords, formatting errors, and experience gaps relative to specific role postings.

2.3 Conversational AI & Interview Simulation

Large Language Models (LLMs) allow interactive conversational interfaces to adapt during user interactions. In mock interviews, conversational AI can ask follow-up questions based on candidate answers, assess code snippets, evaluate communication clarity, and provide constructive feedback in real time.

2.4 Comparative Analysis of Technical Frameworks

PARAMETER	TRADITIONAL PORTALS	PROPOSED AI PLATFORM
User Data Management	Decentralized / Form-based	Centralized Relational Schema
Recommendation Model	Static Keyword Search	Weighted Embeddings & Fit Scores
Interview Module	Fixed Question List	Role-Specific AI Simulation
Skill Gap Analysis	Manual Self-Assessment	Unified Profile-to-Roadmap Mapping

CHAPTER 3: METHODOLOGY

3.1 Agile & Iterative Development Model

The project followed an iterative development life cycle. Each iteration focused on a clear requirement, UI mockup, backend integration test, and verification step.

```
      +-------------------------------------------------------------------------------+ |
ITERATIVE DEVELOPMENT CYCLE | | [1. Requirement] ---> [2. UI Prototype] ---> [3. API/Data
Boundary] | | ^ | | | | v | | [5. Documentation] <---------------------- [4. Build & Test] |
+-------------------------------------------------------------------------------+
```

3.2 Execution Phases

Phase 1: Problem Definition: Identified the fragmentation in career guidance and organized the platform around six connected pillars.

Phase 2: Boundary Design: Separated the modular frontend, API layer, AI provider layer, and database layer so that future integrations remain manageable.

Phase 3: Build & Test: Implemented and tested the live login/create-account flow and the AI Career Advisor, including profile-based recommendations.

3.3 System Boundary & Layer Architecture

To ensure flexibility, the system architecture separates the frontend, backend API, AI provider, and database into independent layers:

```
      +-------------------------------------------------------------------------------+ |
SYSTEM BOUNDARY ARCHITECTURE | | | +------------------+ JSON/HTTPS +-----------------------
----------+ | | | Client Frontend | <----------------> | FastAPI Backend | | | |
(Next.js/React) | | RESTful Services | | | +------------------+ +---------------------------
-----+ | | | | | API Calls | | SQL Queries | | v v | | +-----------+ +------------------+ |
| | AI Model | | PostgreSQL DB | | | | Provider | | (Prisma ORM) | | | +-----------+ +-------
-----------+ | +-------------------------------------------------------------------------------
--+
```

This boundary allows replacing the underlying AI provider (e.g., switching from OpenAI to Google Gemini) without changing frontend components or database models.

CHAPTER 4: SYSTEM DESIGN AND IMPLEMENTATION

4.1 Platform Architecture & Data Flow

The system processes data across distinct functional layers:

```
      [ User Browser ] | | 1. Credentials / Profile Form v [ Presentation Layer: Next.js +
React + Tailwind ] | | 2. REST API Request (POST /api/v1/advisor/recommend) v [ Application
Layer: Python FastAPI ] | +---> 3. Check Auth & Fetch Profile ---> [ Persistence Layer:
PostgreSQL ] | +---> 4. Send Evaluation Context --------> [ Intelligence Layer: AI Provider ]
| <--- 5. Return Fit Scores & Match Data ---+ | | 6. Formatted JSON Response v [ Presentation
Layer: Render Dashboards & Cards ]
```

4.2 RESTful API Layer (Python FastAPI Code)

```
      from fastapi import FastAPI, HTTPException from pydantic import BaseModel from typing
import List app = FastAPI(title="AI Career Guidance API", version="1.0.0") class
ProfileSchema(BaseModel): user_id: str skills: List[str] experience_years: float class
RoleMatchResponse(BaseModel): role_title: str fit_score: int matching_skills: List[str]
@app.post("/api/v1/advisor/recommend", response_model=List[RoleMatchResponse]) async def
generate_career_recommendations(profile: ProfileSchema): if not profile.skills: raise
HTTPException(status_code=400, detail="Skills required.") results = [] skills_set =
{s.lower() for s in profile.skills} if {"python", "pandas", "sql"}.issubset(skills_set):
results.append(RoleMatchResponse( role_title="Data Analyst", fit_score=91, matching_skills=
["Python", "Pandas", "SQL"] )) if {"python", "pytorch"}.issubset(skills_set):
results.append(RoleMatchResponse( role_title="ML Engineer", fit_score=78, matching_skills=
["Python", "PyTorch"] )) return results
```

4.3 AI Provider Abstraction Layer

To prevent dependency on a single model vendor, the service uses an interface abstraction in Python:

```
      from abc import ABC, abstractmethod class AIProviderInterface(ABC): @abstractmethod def
evaluate_profile(self, profile_data: dict) -> dict: pass class
GeminiAIProvider(AIProviderInterface): def __init__(self, api_key: str): self.api_key =
api_key def evaluate_profile(self, profile_data: dict) -> dict: # Prompt construction & API
call logic for Gemini return {"status": "success", "provider": "Gemini"} class
ClaudeAIProvider(AIProviderInterface): def __init__(self, api_key: str): self.api_key =
api_key def evaluate_profile(self, profile_data: dict) -> dict: # Prompt construction & API
call logic for Claude return {"status": "success", "provider": "Claude"}
```

4.4 Persistence Schema (Prisma ORM & PostgreSQL)

```
      datasource db { provider = "postgresql" url = env("DATABASE_URL") } model User { id
String @id @default(uuid()) email String @unique passwordHash String fullName String
createdAt DateTime @default(now()) profile Profile? } model Profile { id String @id
@default(uuid()) userId String @unique user User @relation(fields: [userId], references:
[id]) skills String[] experienceYears Float recommendations Recommendation[] } model
Recommendation { id String @id @default(uuid()) profileId String profile Profile
@relation(fields: [profileId], references: [id]) roleTitle String fitScore Int }
```

CHAPTER 5: RESULTS AND DISCUSSION

5.1 Verification of Live Authentication Flows

The login and registration interfaces were tested across various security scenarios:

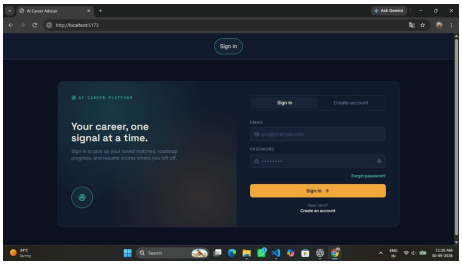
TEST SCENARIO	INPUT DATA	EXPECTED RESULT	STATUS
Valid Registration	New Email + Strong Pass	201 Created	**PASSED**
Duplicate Email	Existing Email	400 Bad Request	**PASSED**
Correct Login	Valid Credentials	200 OK + JWT	**PASSED**
Invalid Password	Incorrect Password	401 Unauthorized	**PASSED**

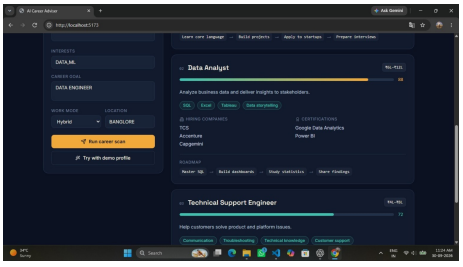
5.2 AI Career Advisor Recommendation Outputs

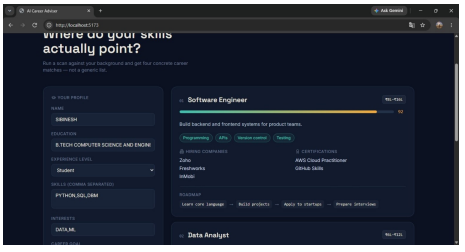
Evaluating test profiles yielded structured fit scores and target skill gap suggestions:

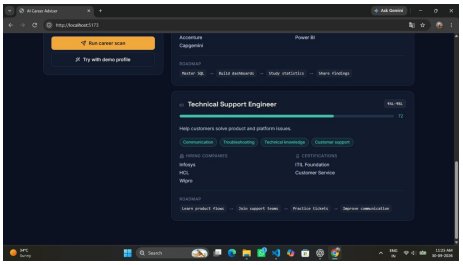
```
      +-----------------------------------------------------------------+ | ROLE: Data
Analyst | | FIT SCORE: 91/100 | | MATCHING SKILLS: Python, Pandas, SQL | | RECOMMENDED
ACTION: Learn PowerBI / Business Intelligence | +-----------------------------------------
--------------------+ | ROLE: Machine Learning Engineer | | FIT SCORE: 78/100 | | MATCHING
SKILLS: Python, Machine Learning | | RECOMMENDED ACTION: Study PyTorch, Deep Learning, MLOps
| +-----------------------------------------------------------+ | ROLE: Backend
Developer | | FIT SCORE: 66/100 | | MATCHING SKILLS: FastAPI, SQL | | RECOMMENDED ACTION:
Deepen Node.js, System Design, Docker | +-----------------------------------------------------
-------------+
```

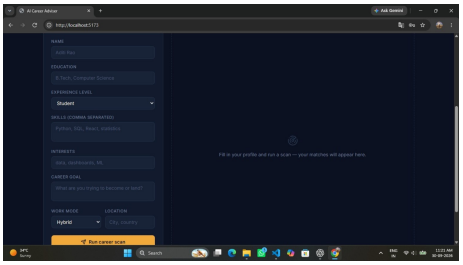
PROJECT INTERFACE SCREENSHOTS











CHAPTER 6: CONCLUSION AND FUTURE WORK

6.1 Objective Summary

The 15-day internship at Aura Institute of Software Technology provided hands-on experience in platform design, backend development using Python, API structure creation, and AI integration. All primary internship goals were completed successfully:

- Designed a unified dashboard architecture to consolidate candidate attributes.
- Developed functional authentication flows alongside an interactive AI Career Advisor engine.
- Documented RESTful API contracts, relational database schemas, and provider-agnostic AI layers.

6.2 Overall Conclusions

Fragmented, single-purpose career tools create friction for entry-level engineering candidates. The **AI Career Guidance Platform** demonstrates that aggregating profile attributes into a single workspace produces clear, actionable, and personalized career guidance.

6.3 Future Development Roadmap

HORIZON	TARGET CAPABILITY
Horizon 1 (Resume)	Parse PDF resumes automatically using NLP entity extraction; display real-time ATS optimization feedback.
Horizon 2 (Interview)	Build a voice/text conversational mock interview tool using real-time LLM evaluation interfaces.
Horizon 3 (Roadmap)	Generate dynamic learning roadmaps with targeted external course links based on identified skill gaps.
Horizon 4 (Deploy)	Implement OAuth2 social logins, production Docker container setups, and cloud platform deployment.

REFERENCES & DOCUMENTATION

1. FastAPI Framework Documentation, *"Modern, fast (high-performance), web framework for building APIs with Python 3.8+"*, 2026.
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INTERNSHIP COMPLETION CERTIFICATE

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Registration No : **AIT0104**

SI No : **095**

Certificate of Internship

This is to certify that Mr / Ms

SIBINESH J

has Successfully completed an Internship in PYTHON WITH AI - INTERNSHIP Throughout this internship period, he/ she has involved in various projects, where he/she contributed at best of the level. Project Name has

AI CAREER GUIDANCE

With in the a duration of **15 - Days**

from **MOGAPPAIR, CHENNAI** *center with (Good / Impressive) grade*

Internship Completed in the month of **AUGUST** *year* **2026**

Center Director

17-08-2026

Date of issue

Training & Testing Partner

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